

Progress Report Sprint 12

- Course Name: CS463 Senior Software Engineering Project III
- Date: 04/26/2026
- Team Name: TaxonBodyMassML
- Team ID: CS.032
- Student Names and Emails:
 - Grant Pasquantonio (pasquang@oregonstate.edu)
 - Hailey Prater (praterh@oregonstate.edu)
 - Anish Chowdary Pendurti (pendurta@oregonstate.edu)

Summary - Outcomes, blockers, and insights

- **UI:** Continued to make informed changes to the UI based off of last term's second-phase interviewing and created the project's landing page
- Started working with project partner to get the hardware we require to host our regressor microservice
- The code for the regressor microservice is at a deployable state. We have added the multi-species query to the microservice, which allows the Frontend to request multiple species simultaneously.

Blockers:

- Our regressor microservice is still not functioning; we have come up with a possible solution, but have not implemented it

Work Log

1. Landing Page

Status: Complete

Owner: Hailey **Co-Owner:** Grant

Description: To comply with a course assignment, we created a 'Landing Page' to supply a potential audience with a summary of the project and its value. This was created as a "More Information" page in the navigation bar.

Source Link: [Landing Page Assignment Submission](#)

Supporting Artifact: Item.01

2. Cosmetic UI Updates

Status: Complete

Owner: Hailey

Description: During my second-phase interviewing, I got a lot of cosmetic feedback mentioning the lack of intrigue and over-use of whitespace on the site. I began to fix the whitespace issue last term by making the modals bigger, but this sprint I tackled both issues at once by adding another graphic and a title on the navigation bar. I also fixed the tutorial pop-ups, which had been muddled after the change in styling from last term.

Source Link: [Spring Term UI Plan](#)

Supporting Artifact: Item.02

3. Multi-species prediction for XGBoost Regressor

Status: In-progress

Owner: Grant

Description: Implemented the API route for a multiple species prediction for the XGBoost Regressor microservice. At this point, the XGBoost Regressor should be in a ready state to be deployed onto server hardware.

Source Link: [Sprint 11 Doc](#)

Supporting Artifacts: Item.03

Notes: The API response exactly matches the single species except it is contained within a JSON array labeled "items".

Risk and Quality Tracking

Open Bug Count: 4 [TaxonBodyMassML Risk Tracker](#)

TOP BUGS	SEVERITY	OWNER(S)	STATUS	LINK	ACTIONS
NON-RESPONSIVE CSS	Low	Hailey	In-progress	primary CSS file on GitHub	Hailey will add in CSS that better accommodates smaller screen sizes (media queries)
CACHED QUESTIONS ARE NOT LOADING	Low	Hailey	In-progress	'More questions' service python file on GitHub	Hailey will debug the API call and JS code and then, if necessary, see if it is an issue with the render deployment
REGRESSOR MICROSERVICE TOO BIG FOR RENDER FREE TIER	High	Grant and Hailey	In-progress	Render deployment status	We will contact the project partner about accessing dedicated hardware with enough on board memory to handle the Regressor Microservice.
NO ERROR MESSAGES FOR UNMATCHED SPECIES	High	Hailey	In-progress	Source of microservice calls	This can only be completed after the previous bug is patched. After we have both functioning microservices, we can

Next Sprint Plan

- We need to fix our regression microservice deployment and get body mass predictions displayed to the front-end. We have decided on a fix revolving around hardware, though we still need to work with Mark on purchasing the tech. This is a large part of the project expectations from our project partner, so this is something we are focusing on immediately.
Priority: High; Links: [2025-10-23-CS.032-Requirements](#) and [Risk Tracker](#)
- Now that the multi-species query is done for the regressor microservice, we must also finish the multi-species query for the lookup-table microservice. This is in line with REQ-003 which indicates that the user must be able to make at least 10 simultaneous queries.
Priority: Med; Link: [2025-10-23-CS.032-Requirements](#) (REQ-003)

Team Reflection

This sprint, we made good progress on all fronts. We were able to get the majority of the framework for a Retrieval Augmented Generation based LLM predictor. We continued working on improving the UI in accordance with feedback from the project partner and made several cosmetic improvements as suggested during second-phase user interviews. We also completed the landing page for the project. Lastly, we were able to complete the multi-species query functionality in the regressor microservice.

The biggest priority for the next sprint is to begin working on a solution for the regressor microservice. We have come to the conclusion that a hardware option is likely the best solution, so we will proceed with that in mind. The goal will be to secure dedicated hardware such as a raspberry pi or similar small computer to run the regressor microservice.

Furthermore, we will continue to work on refining the UI according to feedback from our project partner. Good progress was made on the LLM based microservice, but we will get the predictor in a close-to-completed state by the end of the next sprint. We will continue work on implementing the multi prediction functionality in the lookup table microservice.

Artifacts Appendix

Item.01



The TaxonBodyMassML Project

This website and the predictive algorithms it uses were created as part of a senior capstone project at Oregon State University; here are some of the details!

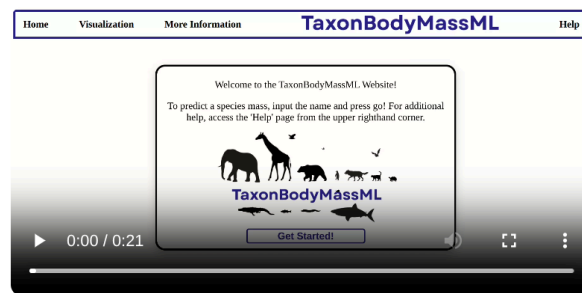
Project Description

Body mass is one of the most powerful predictors of species behavior, interaction, and ecological impact; however, most species on Earth still do not have recorded body mass measurements. Since it is difficult to get a reliable measurement of species mass for unmeasured species, a close estimation would be valuable to ecologists.

Key Features

- Generate mass predictions for species with unmeasured masses
- View associated confidence intervals for each prediction
- Query large datasets of species simultaneously using a .csv

Video Demonstration



How Do I Try It Out?

If you have made it this far, you're just a click away from trying out our project! In the top left of the navigation bar at the top of the screen, simply press the 'Home' button to be taken to the main UI point of TaxonBodyMassML. Go ahead and experiment with our body mass prediction algorithms!

Meet The Team

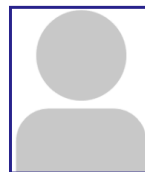
This is the crew that brought the TaxonBodyMassML project to life in the 2025-2026 academic year:



Hailey Prater - Front-End Engineer
GitHub: [praterh](#)



Grant Pasquantonio - Machine Learning Engineer
GitHub: [pasquang](#)



Anish Chowdary Pendurti - Machine Learning Engineer



Mark Novak - Project Partner
GitHub: [marknovak](#)

If you run into any technical issues with the website, please email Hailey at haileyprat59@gmail.com

Fig.01 - This is the landing page for our project. It is reachable through the 'The TaxonBodyMassML Project' tab of the 'More Information' dropdown.

Item.02

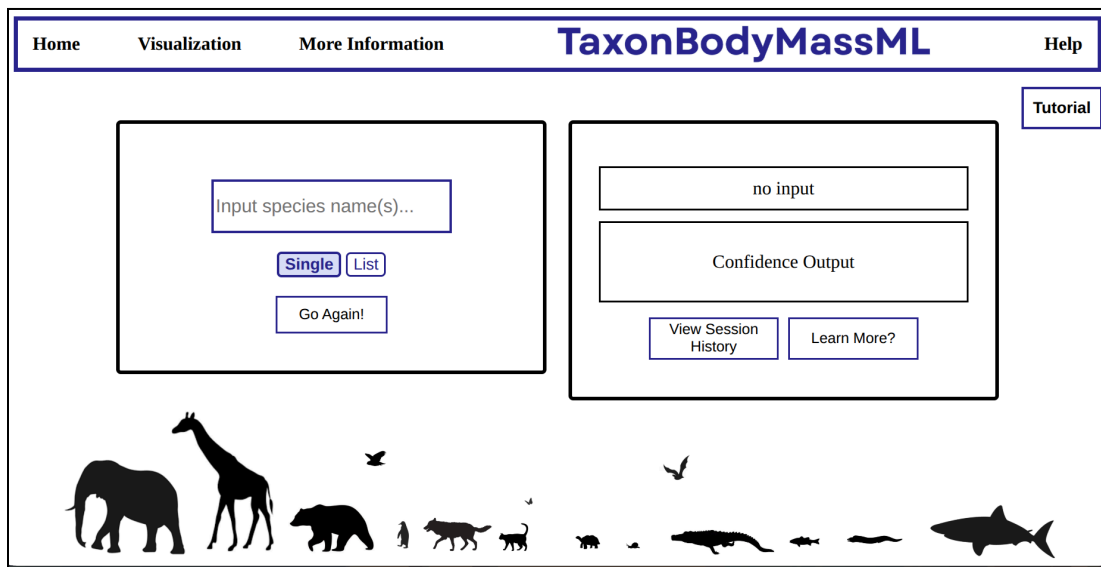


Fig.02 - This depicts two of the 'visual intrigue' elements implemented this week. The graphic along the bottom ties in the logo from the intro modal and the text in the top fills whitespace and keeps the project name prominent.

Item.04

```

Batch Prediction Result:
{
  "items": [
    {
      "confidence": 0.9,
      "lower_bound": 374.6805114746094,
      "prediction": 1693.4993896484375,
      "taxonomy": {
        "class": "Mammalia",
        "family": "Muridae",
        "genus": "Mallomys",
        "kingdom": "Animalia",
        "order": "Rodentia",
        "phylum": "Chordata",
        "species": "Mallomys gunung"
      },
      "upper_bound": 7654.36181640625
    },
    {
      "confidence": 0.9,
      "lower_bound": 10.164824485778809,
      "prediction": 45.94347381591797,
      "taxonomy": {
        "class": "Aves",
        "family": "Cuculidae",
        "genus": "Cacomantis",
        "kingdom": "Animalia",
        "order": "Cuculiformes",
        "phylum": "Chordata",
        "species": "Cacomantis"
      },
      "upper_bound": 207.65756225585938
    },
    {
      "confidence": 0.9,
      "lower_bound": 201.4924774169922,

```

Fig.03 - This is the results of using the multi species query test script with the updated regressor microservice. As can be seen, each response has values for confidence, lower bound, upper bound, and prediction value. The information for each species is contained within a JSON array labeled "items".